Sang Min Lee

Ph.D. Candidate in Artificial Intelligence · Structural & Wind Engineering

Seoul, Republic of Korea

201612445@snu.ac.kr | concrete-sangminlee.github.io

Research Interests

- AI for wind engineering: wind load, pressure, and force estimation; sensor reduction for wind-resistant design
- Learning-based structural assessment: health monitoring, anomaly detection, automated impact-echo evaluation
- Large language models for engineering practice: retrieval-augmented generation and mixture-of-experts for design codes
- Physics-informed and data-driven modeling of structural and material behavior

Education

Ph.D. in Artificial Intelligence (Interdisciplinary Program), Seoul National University — *Seoul, Republic of Korea*
- Dissertation: "Deep Learning Application for Wind-Resistant Design of Building Structure" — *Mar. 2023 - Feb. 2027 (expected)*
- Advisor: Prof. Thomas Kang

M.S. in Architecture and Architectural Engineering, Seoul National University — *Seoul, Republic of Korea*
- Thesis: "Machine Learning Application to Automated Impact-Echo Test for Concrete Structures" — *Mar. 2021 - Feb. 2023*
- Advisor: Prof. Thomas Kang

B.S. in Electrical and Computer Engineering (Double Major), Seoul National University — *Seoul, Republic of Korea*
Sep. 2017 - Feb. 2021

B.S. in Architecture and Architectural Engineering, Seoul National University — *Seoul, Republic of Korea*
Mar. 2016 - Feb. 2021

Science High School for the Gifted, Korea Science Academy of KAIST — *Busan, Republic of Korea*
Mar. 2013 - Feb. 2016

Experience

Graduate Research Assistant, Structural Engineering Laboratory, Seoul National University — *Seoul, Republic of Korea*
- Research projects: — *Mar. 2021 - Present*
  - Hyper-converged Forensic Research Center for Infrastructure (2021~2027, NRF/MSIT)
  - Smart City Innovative Technology Demonstration (2024~2025, KAIA/MOLIT)
  - Outstanding Patent Prototype Development (2023, SNU)
  - AI for Construction (2024, SNU/Sejong Construction)

Academic Reviewer, Wind and Structures (peer-reviewed SCIE journal)
- Volunteer reviewer in the field of wind engineering — *Jan. 2024 - Present*

Visiting Student Researcher, The University of Oklahoma — *Norman, OK, USA*
- School of Civil Engineering and Environmental Sciences & National Weather Center — *Oct. 2025 - Nov. 2025*

Conference Organizing Committee Member, Wind Engineering Institute of Korea — *Seoul, Republic of Korea*
- Organizing committee, 28th WEIK Annual Conference & General Assembly (hosted at SNU) — *Apr. 2025 - May 2025*

Graduate Teaching Assistant, Seoul National University — *Seoul, Republic of Korea*
- Architecture and AI (22S, 23S, 24S) — *Mar. 2021 - Jun. 2024*
  guest lecturer on hands-on deep neural networks
- Introduction to IoT, AI, and Big Data (21S, 21F, 22S, 22F)
  IoT project evaluator
  Arduino UNO
  head TA (22F)
- Creative Engineering Design (21F, 22S)
  IoT project evaluator
  Arduino UNO, ESP32
- IoT and Creative Engineering Design (21F, 22S)
  IoT project evaluator
  Arduino UNO, Teensy, ESP32, STM boards

- Structural Experiment and Material (22S)
  Euler buckling, truss deflection, shake-table seismic test
- Mechanics of Materials for Architectural Engineering (21F)
  fundamentals for structural engineering
- Prestressed Concrete (21F)
  class management for post-tensioned structures
- Computation in Architectural Engineering (21S)
  MATLAB for structural engineering students

Undergraduate Research Assistant, Seoul National University — *Seoul, Republic of Korea*

- HpSE, SNU (2021) — *Dec. 2018 - Feb. 2021*
  wind pressure coefficient prediction for wind-resistant building design
- SNUCEM (2019~2020)
  AHP analysis for improvement of military dining facilities
- AEPRL, SNU (2018~2019)
  IoT device (hardware and software) for indoor surface-condensation detection

Research Engineer (Intern), Chungyeon Architects — *Seoul, Republic of Korea*

- Data analysis for building energy performance evaluation (ECO2, AutoCAD, Python) — *Jul. 2018 - Aug. 2018*

Publications

International Journal Articles

1. **Lee, S. M.**, Alinejad, H., and Kang, T. H.-K. (2026) "Wind force time history reconstruction with reduced pressure taps using deep learning," *Engineering Applications in Artificial Intelligence*. Under review.
2. Yan, S., Gupta, H., Chen, M., Zhu, S., Li, Z., Wen, Y., Zhang, M., Cao, J., Chen, X., **Lee, S. M.**, Kang, T. H.-K., Duan, Q., Sorooshian, S., and Hong, Y. (2026) "Towards a reasoning-based physics-informed framework for efficient parameter specification in earth system modeling," *Science Advances*. Under review.
3. **Lee, S. M.**, Yan, S., Hong, Y., and Kang, T. H.-K. (2026) "Synoptic and non-synoptic wind classification using pseudo-labeling and machine learning with the Oklahoma Mesonet database," *Journal of Wind Engineering and Industrial Aerodynamics*. Under review.
4. **Lee, S. M.**, Hong, J., Choi, H., and Kang, T. H.-K. (2025) "Machine learning assisted method for automated impact-echo testing of concrete structure," *Journal of Nondestructive Evaluation*, 44(4), 121.
5. Lee, Y. I., **Lee, S. M.**, and Kang, T. H.-K. (2025) "Multiclass deep support vector data description for structural health monitoring," *ASCE Journal of Structural Engineering*, 151(12).
6. **Lee, S. M.**, Choi, H.-S., Kim, C., and Kang, T. H.-K. (2025) "Lightweight alternative machine learning model for automating concrete crack image classification," *ACI Structural Journal*, 122(5).

Domestic Journal Articles

1. Ahn, B., Lee, D., **Lee, S. M.**, and Kang, T. H.-K. (2023) "Research on satellite image-based wind speed prediction methodology using neural network," *Journal of Wind Engineering Institute of Korea*, 27(4), 135–142. [In Korean]
2. **Lee, S. M.**, Sung, H. S., and Kang, T. H.-K. (2022) "Comparison of performance for predicting compressive strength of concrete using machine learning," *Journal of the Korea Concrete Institute*, 34(5), 505–513. [In Korean]

International Conference Proceedings

1. **Lee, S. M.** and Kang, T. (2025) "Retrieval augmented generation for assistance of using wind load design codes," *The 10th Asia-Pacific Conference on Wind Engineering (APCWE10)*, Chengdu, China.
2. **Lee, S. M.**, Lee, Y. I., and Kang, T. (2024) "Multi-sphere deep SVDD for system identification of building structure," *The 2024 World Congress on Advances in Civil, Environmental, & Materials Research*, Seoul, Republic of Korea.
3. **Lee, S. M.**, Choi, H.-S., Kim, C., and Kang, T. (2024) "Representation learning for classification of concrete crack images," *The 2024 Structures Congress*, Seoul, Republic of Korea.
4. **Lee, S. M.** and Kang, T. (2022) "Clustering wind pressure tap using dynamic time warping," *8th European-African Conference on Wind Engineering (EACWE2022)*, Bucharest, Romania.
5. **Lee, S. M.** and Kang, T. (2022) "Shear strength prediction of prestressed concrete based on machine learning," *The 2022 Structures Congress*, Seoul, Republic of Korea.
6. **Lee, S. M.** and Kang, T. (2021) "Prediction of wind pressure coefficients on high-rise building façade using LSTM RNN model for sensor reduction," *The 2021 World Congress on Advances in Structural Engineering and Mechanics*, Seoul, Republic of Korea.

Domestic Conference Proceedings

1. **Lee, S. M.** and Kang, T. (2026) "Deep learning-based reconstruction of wind force time history from reduced pressure tap measurements," *The Wind Engineering Institute of Korea 2026 Conference*, Gwangju, Republic of Korea. [In Korean]
2. **Lee, S. M.** and Kang, T. (2026) "Development of edge AI impact echo system for internal defect inspection in concrete structure," *Korea Concrete Institute 2026 Spring Conference*, Changwon, Republic of Korea. [In Korean]
3. Park, H. G., **Lee, S. M.**, and Kang, T. (2026) "Application of impact-echo testing to liner-plate-backed concrete with simulated internal defects in nuclear containment buildings," *Korea Concrete Institute 2026 Spring Conference*, Changwon, Republic of Korea. [In Korean]
4. **Lee, S. M.** and Kang, T. (2025) "Clustering of wind pressure coefficients on concrete building facades using deep learning," *Korea Concrete Institute 2025 Spring Conference*, Jeju, Republic of Korea. [In Korean]
5. Lee, Y. I., **Lee, S. M.**, and Kang, T. (2024) "Structural health monitoring of concrete building using multi-class deep SVDD," *Korea Concrete Institute 2024 Fall Conference*, Sokcho, Republic of Korea. [In Korean]
6. **Lee, S. M.** and Kang, T. (2024) "Framework of large language model for structural design code using mixture of experts," *Korea Concrete Institute 2024 Fall Conference*, Sokcho, Republic of Korea. [In Korean]
7. **Lee, S. M.**, Choi, H.-S., Kim, C., and Kang, T. (2024) "Lightweighting concrete crack image classification machine learning models for on-device AI," *Korea Concrete Institute 2024 Spring Conference*, Gwangju, Republic of Korea. [In Korean]
8. **Lee, S. M.**, Choi, H., and Kang, T. (2023) "Automation of impact-echo test using machine learning," *Korea Concrete Institute 2023 Spring Conference*, Busan, Republic of Korea. [In Korean]
9. **Lee, S. M.**, Sung, H. S., and Kang, T. (2022) "Prediction of concrete compressive strength using machine learning," *Korea Concrete Institute 2022 Spring Conference*, Jeju, Republic of Korea. [In Korean]
10. **Lee, S. M.** and Kang, T. (2021) "Prediction of wind pressure coefficient for concrete building wall using LSTM RNN," *Korea Concrete Institute 2021 Fall Conference*, Gyeongju, Republic of Korea. [In Korean]
11. **Lee, S. M.** and Kang, T. (2021) "Prediction of wind pressure coefficients for concrete building walls using LSTM network," *Korea Concrete Institute 2021 Spring Conference*, Yeosu, Republic of Korea. [In Korean]

Patents

1. **Lee, S. M.** and Kang, T. H.-K. (2025) "AI-based defect detection system inside concrete members," EP 23915029.5 (applied).
2. **Lee, S. M.** and Kang, T. H.-K. (2024) "AI-based defect detection system inside concrete members," US 18/875,721 (applied).
3. **Lee, S. M.** and Kang, T. H.-K. (2023) "AI-based defect detection system inside concrete members," KR 10-2480382 (registered) & PCT/KR2023/006392 (applied).
4. **Lee, S. M.** and Kang, T. H.-K. (2022) "Wind load estimation system based on artificial intelligence," KR 10-2611457 (registered) & PCT/KR2023/021982 (applied).

Honors & Awards

1. **AI Fellowship**, Graduate School of Artificial Intelligence, SNU — *2025 & 2026*
2. **AI Paper Competition Excellence Award**, Graduate School of Artificial Intelligence, SNU — *2025*
3. **Graduate School of Artificial Intelligence Fellowship**, Seoul National University — *2023 & 2024*
4. **Best Teaching Assistant Award**, College of Engineering, SNU — *2022*
5. **Conference Outstanding Paper Award**, Korea Concrete Institute — *2022*
6. **Full Scholarship (SEN Engineering Group)**, Dept. of Architecture & Architectural Eng., SNU — *2021*

Licensure & Certification

1. **Big Data Analysis Engineer (National Technical Qualification)**, Korea Data Agency — *2025*

Professional Membership

1. **Student Member**, American Society of Civil Engineers (ASCE) — *2025–2027*
2. **Student Member**, Wind Engineering Institute of Korea (WEIK) — *2023–2027*
3. **Student Member**, Korea Institute for Structural Maintenance and Inspection (KSMI) — *2023–2027*
4. **Student Member**, Architectural Institute of Korea (AIK) — *2023–2027*
5. **Student Member**, American Concrete Institute (ACI) — *2021–2027*
6. **Student Member**, Korea Concrete Institute (KCI) — *2021–2027*